



VERIFIED CONTACT

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LOCATION
Addis Ababa, ET (Remote/Reloc)

CORE COMPETENCIES

- AI & LLM SYSTEMS
- Multi-Model Routing
 - Prompt Caching & FinOps
 - RAG & Vector Embeddings
 - DSPy & Structured JSON
 - Agentic Workflows
- FULL-STACK & DISTRIBUTED
- TypeScript & Node.js
 - React 19 & Next.js
 - Python & FastAPI
 - PostgreSQL & Redis
 - Docker & CI/CD Pipelines
- COMPUTER VISION & EDGE
- YOLOv8 & Object Tracking
 - ArcFace & Deep ReID
 - PyTorch & ONNX Runtime
 - TensorRT Quantization
- FINTECH & SECURITY
- NIST Post-Quantum (ML-KEM)
 - ISO-8583 Gateway Flows
 - Telebirr & Chapa APIs
 - HMAC-SHA256 Tokenization

ACADEMIC HONORS

B.Sc. Honors in Computer Science

Graduated with Highest Distinction (Top 2%)

Thesis: Real-time Multi-Modal Face & Person ReID with Deep Metric Learning

ABREHAM W. SHIFERAW

SENIOR FULL-STACK & AI SYSTEMS ENGINEER

Senior Engineer specializing in high-throughput AI developer tools, post-quantum cryptographic gateways, and edge computer vision pipelines. Track record building Toka SDK (30%-55% LLM cost reduction via prompt caching & model routing) and production banking middleware. Passionate about performant TypeScript, Python, and open-source systems.

REPRESENTATIVE PRODUCTION SYSTEMS & ARCHITECTURES

Toka AI Developer SDK — Multi-Model FinOps & Routing Engine

CREATOR & LEAD SYSTEMS ARCHITECT | TypeScript, Node.js, Python, Open-Source SDK

- Engineered an enterprise-grade AI proxy routing layer cutting LLM token costs by 30%-55% using intelligent semantic prompt caching and fallback model cascades.
- Unified Gemini, Claude, OpenAI, and local Ollama interfaces into a strictly typed, sub-5ms overhead middleware with automatic retry and rate-limit governance.
- Published comprehensive test harness achieving 99.2% test coverage and adopted by developers in multi-tenant SaaS environments.

VaultX — Post-Quantum Banking & Payment Settlement Gateway

CORE SYSTEMS & CRYPTOGRAPHY ARCHITECT | TypeScript, Node.js, NIST PQC, ISO-8583, Telebirr API

- Architected financial settlement middleware implementing NIST-standardized Post-Quantum Cryptography (ML-KEM-768 key encapsulation and ML-DSA-65 signatures).
- Built double-entry ledger verification engine guaranteeing atomicity, zero double-spend anomalies, and ISO-8583 transaction compliance across Telebirr and Chapa rails.
- Achieved sub-120ms round-trip latency for cryptographic payload verification under concurrent transaction stress tests.

Attendo — Edge Computer Vision Biometric Verification Suite

COMPUTER VISION & DEEP LEARNING ENGINEER | Python, PyTorch, YOLOv8, OSNet, ArcFace, TensorRT

- Built real-time person re-identification (ReID) and facial recognition system processing simultaneous multi-camera video streams at 38+ FPS on edge hardware.
- Integrated ArcFace cosine-margin metric learning with OSNet multi-scale features, achieving 98.4% Top-1 verification accuracy under low-light and partial occlusion.
- Applied FP16 TensorRT quantization reducing inference memory footprint by 58% while maintaining sub-30ms recognition pipelines.

OPEN-SOURCE PACKAGES & CULTURAL TOOLS

Ethiopian Calendar & Ge'ez Numerals Engine (npm packages)

High-precision algorithmic conversion between Gregorian and Ethiopian Ge'ez calendars and numerals with 100% boundary testing. Over 50,000+ developer downloads.

Telegram Mini App Full-Stack Boilerplate

Production-ready React 19 + TypeScript + TON integration toolkit for building Web3-enabled Telegram mini-applications with verified cryptographic authentication.

Bare-Metal Rust Microkernel Research

Custom x86_64 operating system kernel implementing 4-level paging, VGA text drivers, cooperative multitasking scheduler, and zero-allocation interrupt handlers.